Three-Body Cooperative Active Aircraft Protection Guidance

LOS Interceptor Guidance, Sliding-Mode Cooperation, and a Wide-Boundary J_2 Extension

Prajith Saravanan

October 31, 2026

Abstract

We summarize two approaches to active aircraft protection: Ratnoo and Shima's line-of-sight (LOS) interceptor guidance, and Kumar and Mukherjee's cooperative sliding-mode control (SMC) . We then extend the two-way J_2 architecture by replacing its narrow-boundary reaching term with a wider saturation boundary layer. Across five simulation scenarios, the proposed WB J2 law preserves interception while reducing defender control cost by a mean of 11.94% and saturation by 17.40%. We also report an implementation audit and the metrics where the modified law does not help.

1 Problem Setting

An aircraft (A) attacked by a homing missile (M) launches a defender (D). M seeks to hit A while evading D; D seeks to intercept M; and A seeks to survive while, in cooperative formulations, assisting D. We use planar point-mass kinematics with constant speeds and bounded lateral acceleration. For $i \in \{A, D, M\}$,

$$\dot{x}_i = v_i \cos\gamma_i, \qquad \dot{y}_i = v_i \sin\gamma_i, \qquad \dot{\gamma}_i = \frac{a_i}{v_i}, \qquad |a_i| \le a_{i,\max}. \tag{1}$$

r_{ij} and λ_{ij} denote relative range and LOS angle for a pair (i, j). The goal is to place D on the A–M LOS so it intercepts M before M reaches A. Related work includes [3–6]; we follow Ratnoo and Shima [1], then Kumar and Mukherjee [2], then our modification of the latter's two-way J_2 implementation.

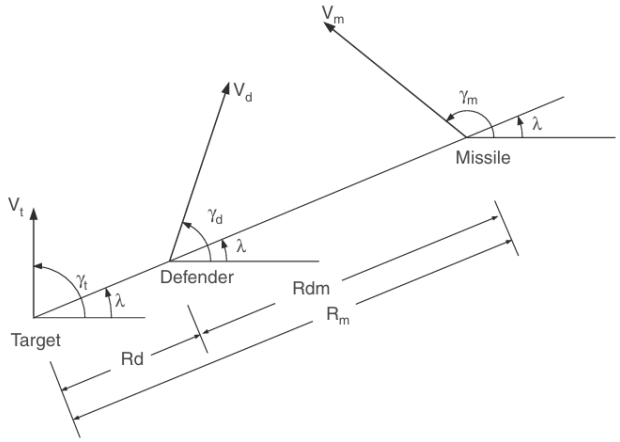


Figure 1: LOS engagement geometry.

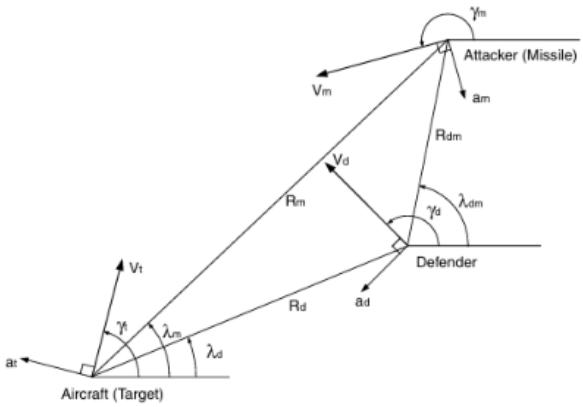


Figure 2: Three-body engagement scenario.

2 LOS Interceptor Guidance: Ratnoo and Shima (2011)

2.1 LOS condition

Let R_m be the attacker–aircraft range and R_d the defender–aircraft range, with $R_d/R_m \in [0, 1]$. Resolving velocities normal to the common LOS gives

$$v_{d,\perp LOS} = \frac{R_d}{R_m} v_{m,\perp LOS} + \left(1 - \frac{R_d}{R_m}\right) v_{t,\perp LOS}. \tag{2}$$

This is attainable when $v_d = v_m$, and interception then holds for arbitrary geometry and attacker maneuver [1]. The required defender normal acceleration is

$$a_{d,\perp LOS} = \frac{R_d}{R_m} a_{m,\perp LOS} + \left(1 - \frac{R_d}{R_m}\right) a_{t,\perp LOS} + 2\dot{\lambda}\left(\dot{R}_d - \frac{R_d}{R_m}\dot{R}_m\right). \tag{3}$$

For a PNG attacker ($N = 3$), a non-maneuvering aircraft, and $v_m, v_d > 2v_t$, $-1 < a_{d,\perp LOS}/a_{m,\perp LOS} < 1$, and the terminal ratio is bounded by about 1/3 under Corollary 1 of [1]. An attacker on a collision course also incurs its maximum evasion penalty when D stays on the A–M LOS ($\lambda_{dm} = \lambda_{am}$).

2.2 CLOS implementation and aircraft cooperation

The paper uses command-to-LOS (CLOS) guidance: the aircraft tracks D and M and uplinks the defender command

$$a_d = KG(s)R_d(\lambda_m - \lambda_d) + a_t + R_d\ddot{\lambda}_m + 2\dot{R}_d\dot{\lambda}_m, \qquad G(s) = \frac{1 + s/a}{1 + s/b},\ b > a, \tag{4}$$

where K is a gain, $G(s)$ a lead network, and a_t a feed-forward for launch-platform motion. The cooperative aircraft maneuver raises the attacker's acceleration demand while keeping D's low:

$$a_t = \begin{cases} -2\dot{R}_d\dot{\lambda}, & |\gamma_t - \lambda| < \pi/2, \\ +2\dot{R}_d\dot{\lambda}, & |\gamma_t - \lambda| \geq \pi/2, \end{cases} \tag{5}$$

initiated when the D–M range falls to half its initial value. The analysis assumes planar motion, constant speeds, no lag and ideal information (including estimates of $\ddot{\lambda}_m$ and $\dot{R}_d$).

2.3 Original simulation study

Our earlier implementation used attacker and defender speeds of 600 m/s and aircraft speed of 200 m/s, comparing CLOS with PNG defenders ($N = 3, 5$) in side-on, head-on and tail-chase engagements, using trajectories, acceleration ratio, miss distance and $J_d = \int_0^{t_f} a_d^2\,\mathrm{d}t$. The trend matched [1]: CLOS kept the LOS geometry with low miss distance and lower effort, while PNG demanded high acceleration and saturated, especially for $N = 3$.

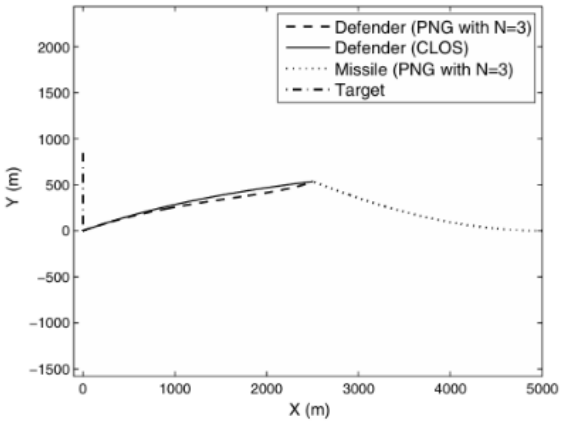


Figure 3: Engagement trajectories, from Ratnoo and Shima [1].

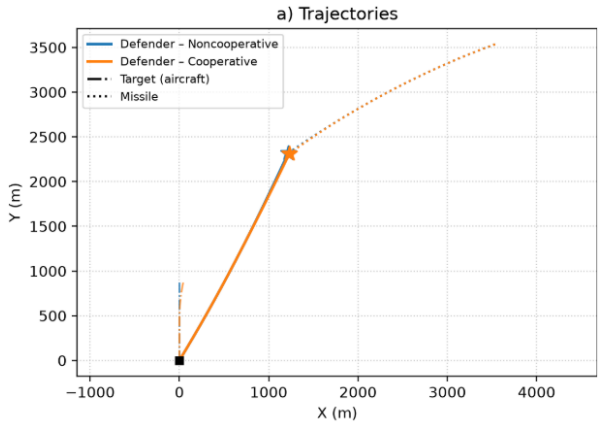


Figure 4: Engagement trajectories, our implementation with aircraft missile LOS of 45 degree

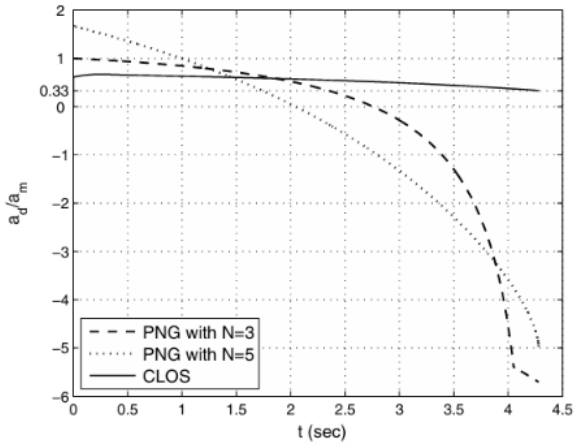


Figure 5: Defender-to-attacker acceleration ratio, from [1].

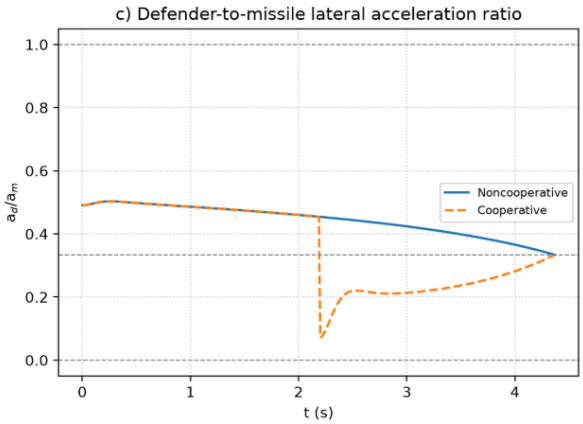


Figure 6: Defender-to-attacker acceleration ratio, our implementation with aircraft missile LOS of 45 degree

3 Nonlinear Cooperative SMC: Kumar and Mukherjee (2021)

3.1 Sliding surface

Kumar and Mukherjee [2] use subscripts a, d, m with $V_d \approx V_m > V_a$ (geometry in Fig. 7).

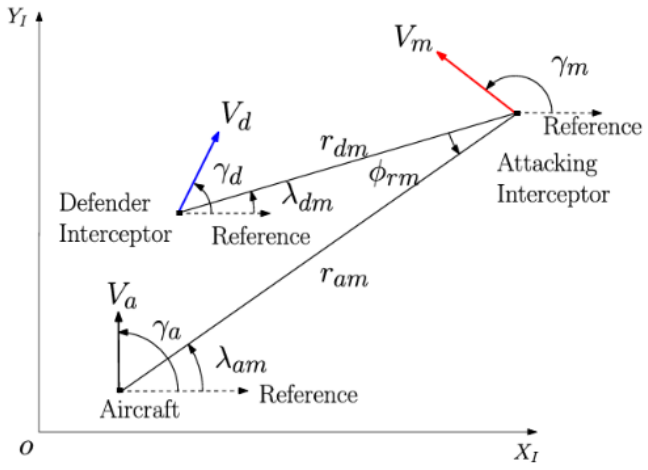


Figure 7: Three-point engagement geometry, from Kumar and Mukherjee [2, Fig. 1, p. 958].

Define $\phi_{rm} = \lambda_{am} - \lambda_{dm}$ (zero when D is on the A–M LOS). Then

$$\ddot{\phi}_{rm} = \Phi_0 - a_1 a_a + a_2 a_d + d, \tag{6}$$

$$\Phi_0 = \frac{2\dot{r}_{dm}\dot{\lambda}_{dm}}{r_{dm}} - \frac{2\dot{r}_{am}\dot{\lambda}_{am}}{r_{am}}, \quad a_1 = \frac{\cos(\gamma_a - \lambda_{am})}{r_{am}}, \quad a_2 = \frac{\cos(\gamma_d - \lambda_{dm})}{r_{dm}}. \tag{7}$$

The unknown attacker-induced term d is bounded by

$$|d| \le d_{\max} = \left(\frac{1}{r_{dm}} + \frac{1}{r_{am}}\right) a_{m,\max}, \tag{8}$$

so only a bound on the attacker's lateral acceleration is needed, not its guidance law. The sliding surface is

$$S = \dot{\phi}_{rm} + \eta \phi_{rm}, \qquad \eta > 0, \tag{9}$$

on which ϕ_{rm} decays exponentially. With $\Phi = \Phi_0 + \eta\dot{\phi}_{rm}$,

$$\dot{S} = \Phi - a_1 a_a + a_2 a_d + d. \tag{10}$$

For $V = S^2/2$ and a reaching law $\dot{S} = -M\ \mathrm{sgn}(S) + d$, $\dot{V} \le -|S|(M - d_{\max}) < 0$ whenever $M > d_{\max}$.

3.2 Type A: one-way cooperation

The aircraft chooses its own maneuver and sends a_a to D as a measured feed-forward input. D uses

$$a_d = a_d^{eq} + a_d^{disc}, \quad a_d^{eq} = \frac{a_1 a_a - \Phi}{a_2}, \quad a_d^{disc} = -\frac{M_1}{\cos(\gamma_d - \lambda_{dm})}\,\mathrm{sgn}(S), \tag{11}$$

giving $\dot{S} = -(M_1/r_{dm})\,\mathrm{sgn}(S) + d$. Reaching holds if (Theorem 1)

$$M_1 > r_{dm} d_{\max} = \left(1 + \frac{r_{dm}}{r_{am}}\right) a_{m,\max}. \tag{12}$$

D alone enforces the sliding mode, and the aircraft feed-forward matters less as r_{dm}/r_{am} shrinks.

3.3 Type B: two-way cooperation and ℓ_p allocation

Here A and D jointly satisfy

$$a_1 a_a - a_2 a_d = F, \qquad F = \Phi + M_2 \operatorname{sgn}(S), \tag{13}$$

which gives $\dot{S} = -M_2 \operatorname{sgn}(S) + d$ and robust reaching for

$$M_2 > d_{\max} = \left(\frac{1}{r_{dm}} + \frac{1}{r_{am}} \right) a_{m,\max}. \tag{14}$$

This is a weaker requirement than (12) for much of the engagement, since the burden is shared. Among the infinitely many solutions of (13), the paper minimizes

$$J_p = \left[\left| \frac{a_a}{w_a} \right|^p + \left| \frac{a_d}{w_d} \right|^p \right]^{1/p}, \qquad (a_d^*, a_a^*) = \arg\min J_p \quad \text{s.t. (13)}, \tag{15}$$

with weights $w_a, w_d > 0$ and $\Gamma = w_a / w_d$. For $p = 2$,

$$a_d^* = -\frac{a_2 F}{a_2^2 + (\Gamma a_1)^2}, \qquad a_a^* = \frac{\Gamma^2 a_1 F}{a_2^2 + (\Gamma a_1)^2}. \tag{16}$$

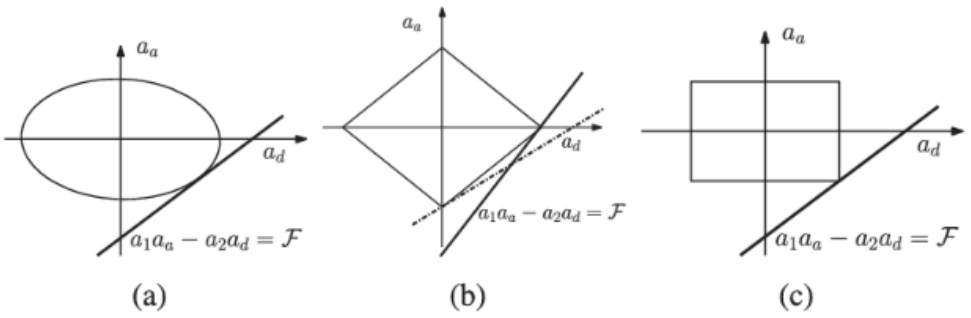


Figure 8: Minimization of the cost function in control space. (a) J_2 (b) J_1 (c) J_p, $p \to \infty$

3.4 Finite-time convergence

[2] also gives a nonsingular terminal sliding surface,

$$\sigma = \phi_{rm} + \frac{1}{\zeta} \dot{\phi}_{rm}^{\xi}, \qquad \xi = \frac{m}{n}, \quad 1 < \xi < 2, \quad m, n \text{ odd}, \ \zeta > 0, \tag{17}$$

where $\xi < 2$ avoids the usual acceleration singularity as $\dot{\phi}_{rm} \to 0$. σ reaches zero in finite time.In simulation, a tanh replaces sgn to reduce chattering, two-way cooperation lets D leave saturation sooner than one-way, and interception succeeds for J_1, J_2, J_4 and J_∞ [2].

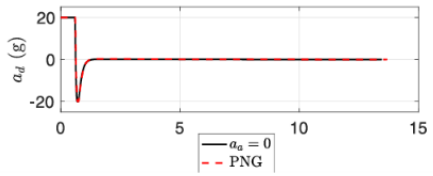


Figure 9: Defender acceleration, Type A

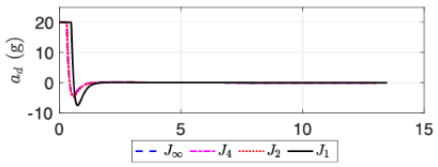


Figure 10: Defender acceleration, Type B

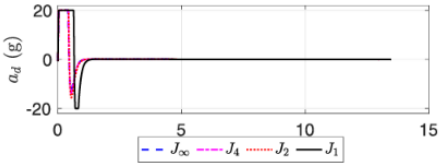


Figure 11: Defender acceleration, Finite time Type B

4 Our Extension: Wide-Boundary J2 Guidance

4.1 Baseline and modification

We keep the engagement model, attacker PNG, sliding surface, disturbance model and $p = 2$ allocation of [2]. Only the reaching term $R(S)$ in $F = B_0 + R(S)$ changes (B_0 collects nominal kinematic terms). We compare

the implemented baseline (hereby called, narrow-boundary J2, NB J2) with wide-boundary J2 (WB J2):

$$R_{\mathrm{NB}}(S) = 0.1 \tanh\left(\frac{S}{0.1}\right), \tag{18}$$

$$R_{\mathrm{WB}}(S) = 0.1\,\mathrm{sat}\left(\frac{S}{1.0}\right), \qquad \mathrm{sat}(z) = \begin{cases} z, & |z| \le 1, \\ \mathrm{sgn}(z), & |z| > 1. \end{cases} \tag{19}$$

Both saturate at magnitude 0.1, but NB has near-origin slope 1.0 while WB has slope 0.1 over a boundary layer ten times wider, so WB reacts less aggressively to small surface errors.

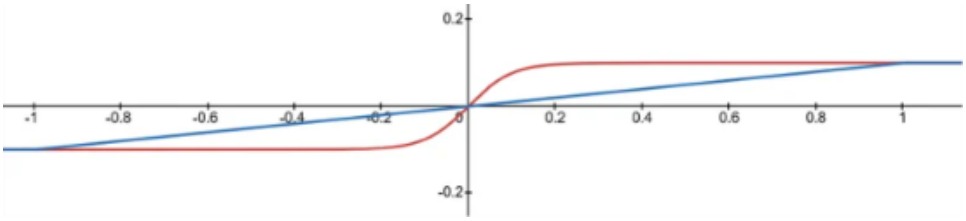


Figure 12: Reaching terms $R_{\mathrm{NB}}(S)$ (tanh) versus $R_{\mathrm{WB}}(S)$ (sat).

4.2 Implementation audit

The implemented plant contains an unknown disturbance term d (representing the attacker's maneuver) that the controller does not explicitly cancel via feed-forward. Therefore, the actual surface dynamics are

$$\dot{S} = d - R(S) + \delta_{\mathrm{act}}, \tag{20}$$

where δ_{act} groups acceleration-clipping and actuator-lag residuals. This residual is common to both laws, so the comparison isolates the reaching-law change but is not an exact realization of $\dot{S} = -R(S)$. Integration is forward Euler with a 1 ms step; events use pre-update geometry.

4.3 Scenarios

We evaluate five engagement scenarios to test robustness. In all cases, disturbances affect only the defender's information; the attacker uses perfect true-state geometry.

- **Ideal:** Baseline performance with no lag and no noise.
- **Lag:** First-order actuator lag on A and D ($\tau = 0.05\,\mathrm{s}$). The attacker has no lag.
- **Noise:** Gaussian noise injected into LOS rates, $\dot{\lambda}^{g}_{ij} = \dot{\lambda}^{true}_{ij} + \nu_{ij}$, where $\nu_{ij} \sim \mathcal{N}(0, \sigma^2_{\dot{\lambda}})$ and $\sigma_{\dot{\lambda}} = 0.002\,\mathrm{rad/s}$. Bearing noise is disabled.
- **Noise + Lag:** Both disturbances active simultaneously (realistic worst-case).
- **Tight Miss:** Endgame interception threshold reduced from 5 m to 1 m.

4.4 Metrics

We evaluate performance using six metrics. We note the desired direction for each:

- **Relative Control Effort:** The ratio of attacker effort to defender effort (higher is better for the defense).
- **Miss Distance:** The minimum separation between the defender and attacker (lower is better).
- **Defender Control Cost:** The total energy expended by the defender over the engagement (lower is better).
- **Saturation Fraction:** The percentage of time the defender's commanded acceleration exceeds its physical limit (lower is better).
- **Chattering (TV):** The total variation of the defender's acceleration command during the final second before impact (lower indicates a smoother, more stable endgame).
- **Terminal Acceleration Ratio:** The magnitude of the defender's terminal acceleration relative to the attacker's (lower is better).

4.5 Results

Both laws intercept in all five scenarios. Table 1 shows the metrics where WB J2 wins in every scenario.

Table 1: Defender control cost $I_{D,2}$ (in $(\mathrm{m/s^2})^2\,\mathrm{s}$) and saturation: NB J2 versus WB J2.

	$I_{D,2}$			$f_{sat,D}$		
Scenario	NB J2	WB J2	Reduction	NB J2	WB J2	Reduction
Ideal	10803	9691.6	10.29%	0.017876	0.015330	14.24%
Lag	10915	9440.0	13.51%	0.021897	0.018716	14.53%
Noise	85404	75754	11.30%	0.051206	0.040088	21.71%
Noise+lag	11167	9570.5	14.30%	0.054213	0.042142	22.27%
Tight miss	10803	9692.2	10.28%	0.017861	0.015318	14.24%
Mean reduction			**11.94%**			**17.40%**

Table 2 gives relative effort and miss distance.

Table 2: Relative control effort (higher is better) and miss distance (lower is better).

	ρ_{ctrl}		d_{miss} (m)	
Scenario	NB J2	WB J2	NB J2	WB J2
Ideal	0.039245	0.040624	4.958	4.7416
Lag	0.042999	0.045480	4.9651	4.7160
Noise	0.0050030	0.0052842	4.9563	4.6152
Noise+lag	0.041270	0.044679	4.7020	4.6521
Tight miss	0.039246	0.040624	0.6925	0.8734

Key Findings:

- **Effort & Accuracy:** WB J2 improves relative control effort across all 5 scenarios (mean +5.33%). It achieves tighter miss distances in 4 of 5 cases. In the `tight_miss` scenario, NB J2 achieves 0.6925 m versus WB J2's 0.8734 m; however, both intercepts successfully satisfy the strict 1 m threshold.
- **Chattering (TV_D):** NB J2 is smoother in perfectly clean environments (`ideal`, `lag`). However, under sensor noise, WB J2 filters the jitter and reduces final-second chattering by 7.38% (`noise`), 7.17% (`noise+lag`), and 16.25% (`tight_miss`).
- **Overall:** WB J2 wins 23 out of 30 valid scenario-metric comparisons, dominating primarily on defender effort and saturation (Table 3).

Table 3: Aggregate WB J2 improvement across the five scenarios.

Metric	Mean WB change	Scenarios favouring WB J2
Relative control effort ρ_{ctrl}	+5.33%	5/5
Miss distance d_{miss}	−1.76%	4/5
Defender control cost $I_{D,2}$	+11.94%	5/5
Defender saturation $f_{sat,D}$	+17.40%	5/5
Chattering indicator TV_D	+0.92%	3/5
Terminal acceleration ratio ρ_{term}	−106.02%	1/5

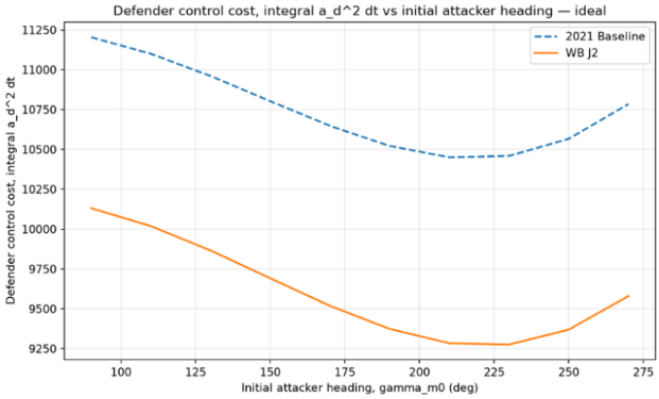


Figure 13: NB J2 versus WB J2 comparison for defender control cost across varying attacker headings.

5 Discussion and Conclusion

Ratnoo and Shima [1] establish the kinematic advantage of keeping the defender on the A–M LOS and give a CLOS implementation for a moving launch platform. Kumar and Mukherjee [2] recast LOS convergence

as nonlinear SMC, with one-way feed-forward or two-way ℓ_p allocation that limits individual acceleration demands without a full-horizon optimal-control solution.

Our WB J2 extension changes only the reaching law. Widening the boundary layer is an effective tuning choice for reducing defender cost and saturation under lag and noisy LOS-rate measurements. It is not a universal improvement: the tight-miss case favors NB J2, and clean-case chattering and terminal-ratio results are mixed.

References

[1] A. Ratnoo and T. Shima, "Line-of-Sight Interceptor Guidance for Defending an Aircraft," *Journal of Guidance, Control, and Dynamics*, vol. 34, no. 2, pp. 522–532, 2011. doi: 10.2514/1.50572.

[2] S. R. Kumar and D. Mukherjee, "Cooperative Active Aircraft Protection Guidance Using Line-of-Sight Approach," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 57, no. 2, pp. 957–967, 2021. doi: 10.1109/TAES.2020.3046328.

[3] L. R. Boyell, "Defending a Moving Target Against Missile or Torpedo Attack," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 12, no. 4, pp. 522–526, 1976. doi: 10.1109/TAES.1976.308338.

[4] N. A. Shneydor, *Missile Guidance and Pursuit*. Horwood Publishing, 1998.

[5] M. Guelman, "Proportional Navigation with a Maneuvering Target," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 8, no. 3, pp. 364–371, 1972. doi: 10.1109/TAES.1972.309520.

[6] V. Shaferman and T. Shima, "Cooperative Multiple Model Adaptive Guidance for an Aircraft Defending Missile," *Journal of Guidance, Control, and Dynamics*, vol. 33, no. 6, pp. 1801–1813, 2010. doi: 10.2514/1.49515.

Acknowledgment: LLMs were used for research assistance and LaTeX formatting. All technical content and results were verified by the author.